

Experiment 10: Implementation of income tax/loan EMI calculator and deploy the same on real devices

PART A

A.1 Aim: To Implementation of income tax/loan EMI calculator and deploy the same on real devices.

A.2 Objectives: To introduce students with various tools like Android Studio, NS2, Wireshark, Cisco packet tracer, WAP supported browser etc.

A.3 Outcome: After successful completion of this experiment students will be able to Implementation of income tax/loan EMI calculator and deploy the same on real devices

A.4 Theory:

SOFTWARE:

- Android Studio
- The Android SDK (Starter Package)
- Gradle
- Java Development Kit (JDK) 5

DESCRIPTION:

15. Open android studio and select new android project .
16. Give project name and select next
17. Choose the android version.
18. Enter the package name. package name must be two word separated by comma and click finish
19. Go to package explorer in the left hand side and select our project.
20. Go to res folder and select layout. Double click the main.xml file 21. Now you can see the Graphics layout window.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent" android:layout_height="match_parent" android:background="#D8D0D0"
    tools:context=".MainActivity">
```

```
<EditText
    android:id="@+id/editText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:layout_marginEnd="8dp"
    android:layout_marginBottom="8dp" android:ems="10"
    android:hint="@string/enter_principle"
    android:inputType="number"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.45"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.203" android:autofillHints=""
/>
```

```
<EditText android:id="@+id/editText2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:layout_marginEnd="8dp"
    android:ems="10"
    android:hint="@string/enter_time"
    android:inputType="number"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.45"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.274"
    android:autofillHints="" />
```

```
<EditText android:id="@+id/editText3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:layout_marginEnd="8dp"
    android:ems="10"
    android:hint="@string/enter_rate_of_interest" android:inputType="number"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.45"
    app:layout_constraintStart_toStartOf="parent"
```

```

app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.386"
android:autofillHints="" />

<Button
android:id="@+id/button"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="8dp"
android:layout_marginTop="8dp"
android:layout_marginEnd="8dp"
android:text="@string/calculate_emi"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.62" />

<TextView
android:id="@+id/textView"
android:layout_width="95dp"
android:layout_height="104dp"
android:layout_marginStart="8dp"
android:layout_marginTop="8dp"
android:layout_marginEnd="8dp"
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.486"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.901" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java

```

package com.example.emicalculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    Button calculate;
    TextView result;
    EditText pe, ne, re;

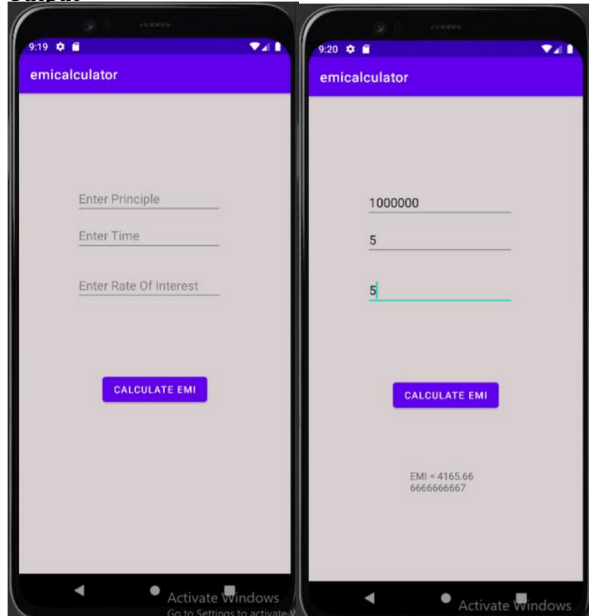
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        calculate = findViewById(R.id.button);
        result = findViewById(R.id.textView);
        pe = findViewById(R.id.editText);
        ne = findViewById(R.id.editText2);
        re = findViewById(R.id.editText3);
        calculate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Double p = Double.parseDouble(pe.getText().toString());
                Double n = Double.parseDouble(ne.getText().toString());
                Double r = Double.parseDouble(re.getText().toString());
            }
        });
    }
}

```

```
        Double rate= (r/1200);  
        Double res = (p * rate * Math.pow(1 + rate, n) / Math.pow(1 + rate, n) - 1);  
result.setText("EMI = "+String.valueOf(res));  
    }  
    });  
}  
}
```

Output



PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

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Class : TE Comps C	Batch: C-3
Date of Experiment: 26.03.25	Date of Submission: 4.04.25
Grade:	

B.1 Software Code written by student/steps:

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:tools="http://schemas.android.com/tools"
    xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="EMI Calculator"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.EMIcalculator">

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

    </application>

</manifest>
```

MainActivity.java

```
package com.example.emicalculator;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Linking UI elements
        EditText etSalary = findViewById(R.id.etSalary);
        Button btnCalculateTax = findViewById(R.id.btnCalculateTax);
        TextView tvTaxResult = findViewById(R.id.tvTaxResult);

        EditText etLoanAmount = findViewById(R.id.etLoanAmount);
        EditText etInterestRate = findViewById(R.id.etInterestRate);
        EditText etTenure = findViewById(R.id.etTenure);
        Button btnCalculateEMI = findViewById(R.id.btnCalculateEMI);
        TextView tvEMIResult = findViewById(R.id.tvEMIResult);

        // Tax Calculation Logic
        btnCalculateTax.setOnClickListener(v -> {
            String salaryStr = etSalary.getText().toString();
            if (!salaryStr.isEmpty()) {
                double salary = Double.parseDouble(salaryStr);
                double tax = calculateTax(salary);
                tvTaxResult.setText("Tax Payable: ₹" + tax);
            } else {
                tvTaxResult.setText("Please enter a valid salary");
            }
        });

        // EMI Calculation Logic
        btnCalculateEMI.setOnClickListener(v -> {
            String loanStr = etLoanAmount.getText().toString();
            String rateStr = etInterestRate.getText().toString();
            String tenureStr = etTenure.getText().toString();

            if (!loanStr.isEmpty() && !rateStr.isEmpty() &&
!tenureStr.isEmpty()) {
                double principal = Double.parseDouble(loanStr);
                double rate = Double.parseDouble(rateStr);
                int tenure = Integer.parseInt(tenureStr);
```

```

        double emi = calculateEMI(principal, rate, tenure);
        tvEMIResult.setText("Monthly EMI: ₹" + emi);
    } else {
        tvEMIResult.setText("Please enter valid inputs");
    }
});
}

// Function to calculate tax
private double calculateTax(double salary) {
    if (salary <= 250000) return 0;
    if (salary <= 500000) return (salary - 250000) * 0.05;
    if (salary <= 1000000) return (250000 * 0.05) + (salary - 500000) *
0.2;
    return (250000 * 0.05) + (500000 * 0.2) + (salary - 1000000) * 0.3;
}

// Function to calculate Loan EMI
private double calculateEMI(double principal, double rate, int years) {
    double monthlyRate = rate / (12 * 100);
    int months = years * 12;
    return (principal * monthlyRate * Math.pow(1 + monthlyRate, months)) /
        (Math.pow(1 + monthlyRate, months) - 1);
}
}

```

Android_main.xml

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    android:orientation="vertical">

    <!-- Income Tax Calculator -->
    <EditText
        android:id="@+id/etSalary"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Annual Salary"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/btnCalculateTax"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Calculate Tax"/>

    <TextView
        android:id="@+id/tvTaxResult"
        android:layout_width="match_parent"

```

```
        android:layout_height="wrap_content"
        android:text="Tax Result will be shown here"
        android:textSize="16sp"/>

<!-- Loan EMI Calculator -->
<EditText
    android:id="@+id/etLoanAmount"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Loan Amount"
    android:inputType="numberDecimal"/>

<EditText
    android:id="@+id/etInterestRate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Interest Rate (%)"
    android:inputType="numberDecimal"/>

<EditText
    android:id="@+id/etTenure"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Tenure (Years)"
    android:inputType="number"/>

<Button
    android:id="@+id/btnCalculateEMI"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Calculate EMI"/>

<TextView
    android:id="@+id/tvEMIResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="EMI Result will be shown here"
    android:textSize="16sp"/>
</LinearLayout>
```


B.2 Input and Output:

EMI Calculator

CALCULATE TAX

Please enter a valid salary

100000

2

3

CALCULATE EMI

Monthly EMI: ₹2864.257873245577

EMI Calculator

CALCULATE TAX

Please enter a valid salary

200000

3

5

CALCULATE EMI

Monthly EMI: ₹3593.738132812697

B.3 Observations and learning:

- Faced issues in `AndroidManifest.xml` (namespace, `android:exported` error).
- Encountered "**Android resource linking failed**" due to incorrect theme references.
- Successfully implemented **EMI & Income Tax calculations** using Java.
- Deployed and tested the app after fixing build errors.

B.4 Conclusion:

- Learned **Android debugging, Gradle fixes, and Manifest configurations**.
- Successfully built and deployed a **functional financial calculator**.
- Future improvements: **Better UI, tax slab integration, and EMI breakdown visuals**.

B.5 Question of Curiosity

1. Explain different steps required to build up this project?

Steps to Build EMI & Tax Calculator App

1 Project Setup:

- Create a new Android project in **Android Studio** with **Java**.

2 UI Design (**activity_main.xml**):

- Add **EditText** (for loan amount, interest rate, etc.), **Button** (to calculate), and **TextView** (to display results).

3 Backend Logic (**MainActivity.java**):

- Implement EMI formula:
$$EMI = \frac{P \times R \times (1+R)^N}{(1+R)^N - 1}$$
- Add logic for **income tax calculation**.

4 Manifest & Permissions (**AndroidManifest.xml**):

- Set **android:exported="true"** for **MainActivity**.

5 Testing & Deployment:

- Run on **Emulator/Real Device** → Fix errors → Deploy successfully